

INTERNATIONAL GCSE **CHEMISTRY**

9202/1

Paper 1

Mark scheme

November 2024

Version: 1.0 Final



2 4 B Y 9 2 0 2 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqa.com](https://www.oxfordaqa.com)

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	(X) freezing	allow evaporation	1	AO1 3.1.1a
	(Y) boiling		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	(initially) temperature increases	allow (then) temperature stays constant (at 36 °C)	1	AO3 3.1.1a
	(then) temperature stays constant (between 2 and 4 minutes)		1	
	(then after 4 minutes) temperature increases		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	36 °C		1	AO3 3.1.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	(at 1 minute) particles are arranged in rows	allow (at 1 minute) particles are closer together	1	AO3 3.1.1a
	or (at 6 minutes) particles are randomly arranged	allow (at 6 minutes) particles are further apart		

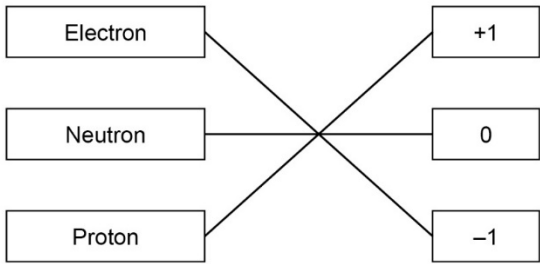
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	melting point (of substance B) is higher than 84 °C	allow (substance B) did not reach its melting point	1	AO3 3.1.1a

Total Question 1		8
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	noble gases		1	AO1 3.1.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	nucleus		1	AO1 3.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	Sub-atomic particle  do not accept more than one line from a box on the left	Charge allow 1 mark if one or two correct	2	AO1 3.1.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	electron		1	AO1 3.1.2j

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	(neon) atoms have stable arrangements of electrons	allow shell for energy level allow (neon) atoms have a full outer energy level	1	AO1 3.1.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	(different) number of energy levels	allow shell for energy level	1	AO1 × 1 AO2 × 1 3.1.2i 3.1.3c
		allow different (total) number of electrons		
		allow (the electronic structure of) helium is 2 and neon is 2,8	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	isotopes		1	AO1 3.1.2g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	(mass of neon =) 0.42×20 = 8.4 (g)		1	AO2 3.6.2a
			1	

Total Question 2		11
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	copper is less reactive than carbon		1	AO2 3.3.1.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	2		1	AO2 3.6.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	(substance) copper oxide	ignore reference to electron gain	1	AO2 3.3.1.1d 3.8.4a
	(reason) (copper oxide) loses oxygen		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(relative formula mass =) $(63.5 \times 2) + 16$		1	AO2 3.6.2a
	= 143		1	

Question	Answers			Extra information	Mark	AO/ Spec. Ref.
03.5		Cu	N	O		
	(moles =)	$\frac{12.7}{63.5}$	$\frac{5.6}{14}$	$\frac{19.2}{16}$		AO2 3.6.2c
		= 0.2	= 0.4	= 1.2		
		$\frac{0.2}{0.2} = 1$	$\frac{0.4}{0.2} = 2$	$\frac{1.2}{0.2} = 6$	allow correct use of incorrectly determined number of moles	
		CuN ₂ O ₆			allow correct use of incorrectly determined number of moles	

Total Question 3	10
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	measuring cylinder	allow pipette / burette	1	AO4 3.9.1a 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	any two from: • volume of water • mass of solid • time of one minute		2	AO4 3.9.1a 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	plastic is a better insulator or (with plastic cup) less energy transfer	allow (with plastic cup) less heat transfer	1	AO4 3.9.1a 3.9.2c

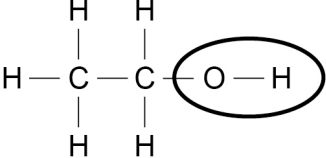
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	0.1 (°C)		1	AO3 3.9.1b,c 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	temperature increases		1	AO3 3.9.1b 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	energy needed to break bonds		1	AO1
	energy released when bonds form		1	3.9.2e 3.9.2g
	(so) more energy taken in than given out		1	

Total Question 4		9
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1			1	AO1 3.10.3.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	(a water bath is easier) to control the temperature		1	AO4 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	35 °C < optimum temperature < 45 °C		1	AO3 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	(conversion) $\left(\frac{210}{60}\right) = 3.5$ (minutes)		1	AO2 3.10.1.2f
	(volume of carbon dioxide =) 3.5×1.5	allow incorrect / no conversion of time	1	
	= 5.25 (cm ³)		1	
	alternative approach			
	(volume of carbon dioxide produced per second = $\frac{1.5}{60}$ =) 0.025 (cm ³ /s) (1)	allow correct use of incorrectly determined volume produced		
	(volume produced in 210 s =) 0.025×210 (1)			
	5.25 (cm ³) (1)			

Question	Answers	Mark	AO/ Spec. Ref.
05.5	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO1 × 4 AO3 × 2 3.10.1.2f
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4	
	Level 1: Relevant points are made. They are not logically linked.	1–2	
	No relevant content	0	
	Indicative content: - sugar cane is a renewable resource but crude oil is a non-renewable resource and may run out - fermentation uses low temperatures so has low energy costs - fermentation uses less energy so less carbon dioxide emissions from burning fossil fuels - hydration needs fractional distillation and cracking so has high energy costs - hydration uses high temperature and pressure so has high energy costs - hydration needs fewer workers so has lower labour costs / greater automation - fermentation needs more workers so has higher labour costs - fermentation is a batch process so rate of production of ethanol is lower - hydration is continuous process so rate of production of ethanol is higher - fermentation has a slow rate of reaction - hydration has a higher rate of reaction - hydration produces higher concentration of ethanol so less energy is used to separate into pure ethanol - judgement		

Total Question 5		12
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	some (oxygen) escaped	allow the student did not add stopper quickly enough allow some (oxygen) dissolved in the water	1	AO4 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	(rate) decreases		1	AO3 3.8.1a
	(rate decreases) more slowly as time increases	allow (rate is) fastest at the beginning	1	
	(rate) becomes zero at 240 seconds	allow (rate) becomes zero when 50cm ³ of gas collected	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	concentration of hydrogen peroxide decreases (as reaction progresses)	allow fewer particles in the same volume of hydrogen peroxide (as reaction progresses)	1	AO2 3.8.1a,e
	(so) less frequent collisions		1	
	(then) reaction stops because hydrogen peroxide is used up		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(observation with metal oxide A) bubbles are produced faster or bubbles stop sooner	allow converse for metal oxide B	1	AO2 × 2 AO3 × 1 3.8.1g 3.9.2h
	(as) the reaction with metal oxide A catalyst is faster		1	
	(because) metal oxide A catalyst provides (a route with) a lower activation energy	MP3 is dependent on MP2 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	(potassium iodide) is unchanged at the end or (potassium iodide) is not used up	allow (potassium iodide) is not in the overall reaction equation	1	AO2 3.8.1g 3.9.2h
	(potassium iodide) provides an alternative reaction route		1	

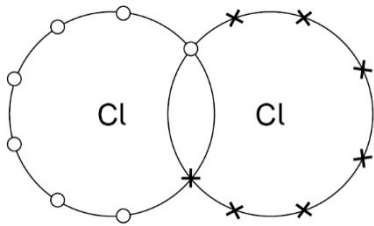
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	(bond breaking) $((463 \times 4) + (2 \times 146) =) 2144$		1	AO2 3.9.2e
	(bond making) $((463 \times 4) + 496 =) 2348$		1	
	(energy change = $2144 - 2348$) = (-204 (kJ/mol))		1	

Total Question 6		15
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	covalent		1	AO1 3.2.1g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	the molecules have no overall electric charge	allow there are no free ions allow there are no delocalised electrons	1	AO1 3.2.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	bonded pair of electrons in the overlap 6 non-bonding electrons on outer energy level of each chlorine atom	allow any combination of x, o, e ⁽⁻⁾ , • for electrons an answer of 	1	AO1 3.2.1g
			1	

scores **2** marks

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	(damp blue litmus paper) turns red		1	AO1 3.4.2e
	(then) turns white or (then) bleaches		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	(in chlorine) the attraction between the outer electrons / shell and the nucleus increases	allow energy level for shell allow (in chlorine) shielding between the outer electrons / shell and the nucleus decreases	1	AO1 3.7.1f
	(so) an electron is gained more easily		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	bromine	allow Br ₂	1	AO2 3.7.1e
	potassium chloride	allow KCl	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.7	(number of moles of Fe = $\frac{22.4}{56}$ =) 0.40		1	AO2 3.6.1d 3.6.4d
	(number of moles of Cl ₂ = $\frac{0.40}{2} \times 3$ =) 0.60	allow correct use of incorrectly determined number of moles of iron	1	
	(conversion 24 dm ³ =) 24 000 cm ³		1	
	(volume of Cl ₂ =) 0.60 × 24 000	allow correct use of incorrectly determined number of moles of chlorine	1	
	= 14 400 (cm ³)	allow correct use of incorrect / no conversion of volume	1	
	alternative method			
	(number of moles of Fe = $\frac{22.4}{56}$ =) 0.40 (1)			
	(number of moles of Cl ₂ = $\frac{0.40}{2} \times 3$ =) 0.60 (1)	allow correct use of incorrectly determined number of moles of iron		
	(volume of Cl ₂ = 0.60 × 24 =) 14.4 dm ³ (1)	allow correct use of incorrectly determined number of moles of chlorine		
	conversion = 14.4 dm ³ × 1000 (1)	allow correct conversion of an incorrectly determined volume		
	= 14 400 (cm ³) (1)			

Total Question 7		15
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	passing an electric current through (a substance)		1	AO1 3.3.2b
	to break down the substance (into elements)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	(electrolysis) uses large amounts of energy		1	AO1 3.3.2i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	ions are free to move		1	AO1 3.2.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	lowers the melting point (of the mixture)		1	AO1 3.3.2i

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	(the positive electrode is) made of carbon		1	AO1 3.3.2i
	(which) reacts with oxygen		1	
	to produce carbon dioxide		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	allow 1 mark for $\text{Al}^{3+} + \text{e}^- \rightarrow \text{Al}$ with no / incorrect balancing numbers	2	AO2 3.3.2f

Total Question 8		10
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